ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Naval Warfare Officer (NWO)
(00207.02, 00207.5, 00207.6, 00207.13)

Command Development Course

National Qualifications and MITE Codes:

- National Qualification Code: Nil
- CDC Residential: 122634
- CDC Sea Phase: 122636

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RECORD OF AMENDMENTS

1. The "Authority" reference indicates approval of this QSP document. Authority for chapters 1 and 2 of this QSP reside with the Commander Naval Personnel and Training Group (CNPTG). The authority for chapters 3 of this QSP Commanding Officer Naval Training Development Centre (Pacific) (CO NTDC (P)), who is both the Training Authority (TA) and the Curriculum Control Authority (CCA).

2. The approval date appears in the bottom right corner of every page (the change date shall be updated in the footer). The Revision column is intended to provide a short indication of where the change(s) was/were made. The history of amendments to previous QSPs is detailed in Annex J "Course History" and shall henceforth be updated with any changes to this version of the QSP.

Table 1 – Record of Amendments

REVISION	APPROVAL DATE	AUTHORITY
First Draft	Jan 2010	DMTE
First Revision – incorporation of lessons learned from first 4 courses, inclusion of ORCA Sea Phase in QSP	Jan 2012	NOTC
Second Revision – introduction of Navigation PC, latest amendments to ORCA Sea-phase	Jan 2014	NOTC
Third Revision – amendment to EO 002.03 (addition of shiphandling case studies)	Jun 2014	NOTC
Fourth Revision - amendment following removal of Command Board from RCN Command Qualification Process	December 2020	CNPTG

FOREWORD

1. This training document is developed under the authority of the Assistant Chief of Naval Staff (Personnel and Training) A/CNS (P&T), and approved by Commander Naval Personnel and Training Group (CNPTG) in accordance with the following references:

    1. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education System (CFITES);
    2. A-PD-055-002/PP-003 Canadian Forces Officer General Specification dated December 2020; and
    3. NPTORD 4500-22 Naval Personnel Training Order – Qualification Standard & Plan.

2. This qualification standard and plan (QSP) publication is effective upon receipt and supersedes CDC QSP revision 2.3, dated Sep 2014.

3. Performance Objectives (POs) and Enabling Objectives (EOs): POs and EOs which specify the standards that students must achieve to become qualified are listed in the Table of Contents and are explained fully in Chapter 3 and Annex K Part 2 Activity and Assessment Plan (AAP).

4. Assessment Details: Assessment details for all Enabling Checks (ECs), Critical Enabling Checks (CECs) and Performance Checks (PCs) can be found in Chapter 2 and 3 of this QSP, Annex K Part 1 - Assessments Overview, Part 2 - Activity and Assessment Plan (AAP) and Part 3 - Assessment Checklists.

5. Activity and Assessment Plan: The Activity and Assessment Plans (AAP) for each PO are located in Annex K Part 2 – AAP.

6. Assessment Checklists: Checklists for all assessments are located in Annex K Part 3.

7. Amendments: Suggestions for amendments may be initiated by anyone and must be forwarded through the normal chain of command summarizing all requested amendments in writing to the applicable approving authority. IAW NPTORD 4500-22 Table 6.1, minor amendments to Chapter 2 and 3 are to be forwarded to course control document managers (CCDM) at NTDC (P) and major amendments and final approval for Chapter 1 amendments are to be submitted to the CNPTG.

8. Description of Need: To ensure that ship commanding officers are prepared for the accountabilities, responsibilities and authorities (ARAs) associated with Sea Command.

9. Scope: The major areas covered by this course are:

    PO 001 – Apply knowledge of Command and Control Theory of Warships;

    PO 002 – Apply professional mariner skills for warship command;

    PO 003 – Apply maritime law, use of force, and tactical mission/deployment planning considerations; and

    PO 004 – Apply command level management theory for warship command.

10. Method of Achieving Objectives: The objectives of the course are met by a combination of theoretical and experiential training and performance-based assessments in the classroom, in simulators, and at sea.

11. This QSP shall be used by NFS(P) as the primary authority governing development, conduct, program evaluation and validation of Command Development Course (CDC).

12. This QSP must be used in conjunction with Occupation Specialty Specifications located in A-PD-055-003/PQ-001.

13. The designated TE for this training is:

Table 2 - Training Establishments

TE	Phase / Module	MITE Course ID	NQual
NFS(P)	Residential Academic Phase	122634	N/A
NFS(P)	Sea Phase	122636	N/A

CHAPTER 1 - **TRAINING MANAGEMENT DETAILS**

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. The aim of this qualification standard and plan (QSP) is to establish the training standard for the development of NWO Command-level knowledge and competencies necessary to achieve a Command Part I Certification. This QSP is designed to assist training establishments in developing, conducting, evaluating, and validating the training (NAVGEN 014/19).

2. In accordance with NAVORD 4500-30 the Command Part I Certification confirms that the officer has demonstrated a sufficient baseline of professional knowledge and competency that he or she will be ready, upon further at-sea experience, to qualify for Sea Command (Command Certification Part II).

3. In 2020, the command qualification process underwent significant change. In accordance with NAVGEN 014/19, From 1 April 2020, the surface command qualification will be granted to all NWO once they have completed 18 months of ORO service in a sea-going position following the ORO course and CDC, unless a fleet commander determines no potential exists. ORO who feel that their performance and/or experience warrant qualification sooner may apply to their respective fleet commanders for early qualification after 6 months experience as an ORO as long as they have completed CDC. For reservists, the requirement will be CDC and 12 months leadership at sea (ORCA OIC or MCDV XO).

COURSE DURATION

4. Course duration is calculated in accordance with NPTORD 4500-22 Annex B. Detailed sequencing and timings are found at Annex G of this QSP. Table 3 below, details the overall estimated and proposed allocation of timing for this course. The estimation is based on the NPTORD 4500-22 direction for allocation of physical training and administrative time. The proposed allocation is based on CDC instructional staff experience from past course schedules.

Table 3 - Course Duration

Training Days Required	**Actual**
Actual Training Time (ATT)	41 days
Performance Checks (PCs) + Enabling Checks (ECs)	11 days
Training Administrative Time (TAT)	1 day
Total Training Days = (ATT) + (PC & ECs) + (TAT)	**53 Training Days**

RESOURCES

5. Required Course Staffing: Table 4 below, details key personnel required to conduct the training:

Table 4 - Required Course Staffing

Required Course Staffing					
Position	Minimum Rank	Qualifications Required	Qty	Remarks	PC/EC, PO/EO
Sr Mentors	Capt(N) /Civi	NWO (MOSID 00207), AEEG	2	Required for through-course instruction and valuation.	All PCs All ECs All POs All EOs
Course Training Administration Officer (CTAO)	LCdr /Civi	NWO (MOSID 00207), AEEG	1	Responsible for all course training, administration, and where required, assessment.	All PCs All ECs All POs All EOs
Current or Post Command CO / (Reg/ResF)	Cdr/LCdr	NWO (MOSID 00207), AEEG	2	Required for final evaluation during ORCA Sea Phase. Post-MCDV Command acceptable. (CFTPO)	PO 001 PO 002D
NABS Bridge Team	NCdt	NWO III min (MOSID 00207)	2	OOW (CFTPO)	EC 002A PC 002B EO 2.02 PC 002C EO 2.03
Helmsman	OS	NETP (ATR)	3	Helmsman (CFTPO)	EC 002A PC 002B EO 2.02 PC 002C EO 2.03

6. Critical Training Resources: Table 5 below, details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timings and sequence found in Annex G and the provision of additional training resources required in Annex H of this QSP.

Table 5 - Critical Training Resources

Critical Training Resources	Training Days	PO/EO
DWAN Laptop (each student) which includes: -WECDIS, NavPac, Rules Master Pro, Total Tides, ADC, MS Office Suite	FC	All

Critical Training Resources	Training Days	PO/EO
O-365 ECN Accounts (each trainee and instructor)	FC	All
CDC Classroom	FC	All
Smart Board	FC	All
DWAN Desktop connected to Smart Board	FC	All
DWAN drops (each student)	FC	All
Colour Printer	FC	All
Navigation and Bridge Simulator (NABS or equivalent)	FC	All
Naval Training Vessel (Orca Class PCT or equivalent)	See Table 6	All
Chart and Publication Outfit	FC	All
USB Thumb Drive (Protected A) (each trainee)	FC	All
Selected and relevant publications as per Annex A or otherwise articulated by CDC instructional staff on a "as required basis."	FC	All

7. Critical Training Resources – Afloat: Tables 6 below detail critical training requirements during the sea phases for this course. See notes following the table for amplification and clarification

Table 6 - Critical Training Resources ORCA Sea Phase

ORCA Sea Phase					
Event	**Platform**	**Days at Sea**	**Resources Based On:**		
			8 Students	**10 - 16 Students**	**17-20 Students**
Assessed	Orca PCT	10	1 Orca PCT 1 CO	2 Orca PCTs 2 COs	3 Orca PCTs 3 COs

COURSE CAPACITY

8. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (Instructor to Learner Ratios).

a. Maximum – 20 students;
b. Minimum – 8 students; and
c. Optimum – 16 students.

COURSE LOADING

9. Loading: The authority to identify candidates and prioritize loading, rests with D Mil C 2 in conjunction with NAVRESHQ. Course loading message to be generated by NPTG HQ.

PREREQUISITES

10. The loading authority must ensure that members have the associated prerequisites listed in table 7 below, before being selected for this course:

Table 7 - Prerequisites for Loading on Course

Regular Force NWO - SSC	Regular Force NWO - SS	Reserve Force NWO
Medically fit sea	Medically fit sea	Medically fit sea
Qualified BWK	Qualified BWK	Qualified BWK
Qualified NOPQ	Qualified Sub-NOPQ	N/A
Qualified ORO	Qualified SMORO	Qualified COPS (I, II, III)
D Mil C 2 Prioritization	CANSUBFOR Prioritization	CO PCC (Q) Prioritization

SCHEDULING

11. Scheduling: Each session of this course shall be scheduled IAW NAVORD 4500-0 Royal Canadian Navy Individual and Collective/Operational Training Policy, by the training establishment conducting the training. The length of the course is to be based on the course duration (see Table 3 above), training breaks, training system holiday, and weekends.

RESERVE FORCE CLASS A PARTICIPATION IN CDC

12. The preferred option for Class A attendance, on the CDC, is to complete the full ten week course in its entirety in one sitting. Understanding the unique employment restrictions faced by Class A reservists, the course may be broken into shorter, residential blocks, completed over a twenty-four month period. As an example, a student may complete the first three weeks of CDC during the Spring serial, the second three week session during the Fall serial, and the final four week session (including the sea phase) in the following Spring serial.

13. The blocks chosen by the student will have to coincide with already scheduled courses. Given the relatively small numbers anticipated, and the unique needs of individual Class A reservists, the length of individual blocks can be flexible but must ensure that on completion, the student has been exposed to all of the course content, including the ORCA sea phase. Prior to commencing a block attendance approach to CDC, a complete training plan must be proposed by the student, communicated to

NFS(P) Venture by their parent unit, which must include refresher training in between blocks as part of that training plan.

MITE

14. MITE Qualification: Two qualifications are provided through the successful completion of this course. Below is the breakdown which indicates the MITE codes associated with each phase of training:

a. CDC Residential Academic Phase **MITE Code 122634**

i. This is awarded upon successful completion of all PCs listed in Chapter 2, Table 8 up to but not including the two PCs represented by the Sea Phase.

b. CDC Sea Phase **MITE Code 122636**

ii. This is awarded upon successful completion of both Sea Phase PCs (namely 001 and 002D).

Note: (Academic MITE) + (Sea Phase MITE) + (+/- 18 months EQ IAW SPCA Process Requirements) = **Surface Command Qualification MITE Code 122637.**

COURSE REPORTING

15. In order for candidates to pass this course and be granted the Command Qualification Part I, they must successfully pass all PCs and complete all ECs. Upon passing this course, students shall be provided a Course Report (CF 377). These are to be completed IAW DAOD 5031-9 - Course Reporting and Certificates. The CF 377 is intended to be a summary of the student's course performance and a prescription for further development and mentoring of the student by their Commanding Officer in preparation for selection to a sea command appointment (i.e. XO, CO).

16. Unsuccessful trainees shall only receive a Course Report (CF377).

SPECIAL INSTRUCTIONS

17. Foreign students that are permitted to be course loaded. Office of interest in the loading of foreign students, resides with D Nav P&T, NDHQ Ottawa.

18. The CF377 for foreign students will indicate course completion of CDC, but will not award a Canadian-specific Part I Command Qualification.

CHAPTER 2 - **ASSESSMENT**

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE NAVAL TRAINING DEVELOPMENT CENTRE (PACIFIC)

GENERAL

1. The first objective of this course is to establish and confirm professional knowledge of selected officers. The second objective is to produce officers with the requisite mariner and leadership skills needed to successfully command at sea.

2. The purpose of the assessment process is to:

   a. determine whether the student has met the standard of knowledge and competency to be awarded a Command Part I certification,

   b. evaluate the students' potential for future appointment to sea command; and

   c. provide a prescription for the officer's further development.

3. The general sequence of instruction/evaluation is:

   a. development and examination of a sufficient knowledge base;

   b. classroom study and problem solving exercises;

   c. practical/experiential learning in a synthetic learning environment, with individual assessment;

   d. practical/experiential learning in a realistic learning environment with individual assessment; and

   e. summative practical examination of competency in a realistic learning environment.

PROGRESS MONITORING

Ref: CFITES Vol. 6 – Conduct of Instruction Programmes

4. The officer will receive feedback on all aspects of performance – including all ECs and PCs. A mid-course interview with instructor is to be provided.

5. Student progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning, so that content is retained.

6. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 "Conduct of Instruction Programmes".

7. Continuous monitoring of student progress is required to provide the following:

a. Early warning of trainee difficulties;
b. Feedback on the effectiveness of training;
c. Information for any required TRBs ; and
d. Information to assist in completion of the CF 377 Course Report.

8. Students experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.

9. Instructional staff shall maintain individual trainee files, including the results of all:

a. Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and
b. Summative assessment material (PCs, CECs).

ASSESSMENT: ENABLING AND PERFORMANCE CHECKS (EC AND PC)

Ref: CFITES Vol. 7 - Evaluation of Learners.

10. Students must successfully complete all POs to pass this course. Performance Checks (PCs), Critical Enabling Checks (CECs) and Enabling Checks (ECs) are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification. Annex K articulates the requirements for each PO/EC.

11. Enabling Checks (ECs): ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to both instructional staff and trainees.

12. In the event of an EC failure, the trainee will be counselled and given remedial training and/or additional homework, as appropriate.

13. Critical Enabling Checks (CECs): CECs are summative assessments at the end of an EO which are considered critical as they represent the core mariner skills which are fundamental in the safe execution of Command at sea.

14. Performance Checks (PCs): PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

15. PCs will be conducted after formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO.

16. Failure of PCs will result in an incomplete designation for that PC and the course. ECs, individually, will not affect the awarding of a Command Part I certification but failure of a majority of CECs, may result in a TRB (Chaired by Cmdt NFS(P) or their delegate) to determine if the officer should proceed to sea for PC 001 and PC 002D – ORCA Sea Phase.

Table 8 - Requirements for Qualification

Performance Objective	
PO 001	Apply knowledge of Command and Control of Warships
PO 002	Apply professional mariner skills for warship command
PO 003	Apply maritime law, use of force, and tactical mission/deployment planning considerations
PO 004	Apply command level management theory for warship command

PHASES

17. Academics and NABS Phase: – General Concept: The Classroom and NABS phase is considered as one because the training consists of classroom study and discussion interspersed with practical application in NABS. The Classroom consists of worksheets, tabletops and discussion about skills which are then practiced and developed in NABS. The NABs sessions are distributed and interspersed throughout the classroom so as to match incremental learning in the classroom with practice in NABS. This Phase becomes incrementally more challenging as the training prepares the students for the Orca Sea Phase;

18. Orca Sea Phases: This constitutes the integration of all prior learning on the course in a practical setting at sea, and the final PC for the Course, PC 001. Although there is a learning element to the phase in ORCA Class competencies and handling (PC 002D), the Assessing officer, a current or recent warship captain, ultimately assesses the

student on their judgement and capacity to command at sea. This is a subjective assessment, whose criteria can be boiled down to whether the assessing officer has confidence that the student can safely exercise command at sea of an ORCA Class vessel in all conditions (day/night, weather etc) .

UNSATISFACTORY TRAINING PROGRESS

19. In the event an instructor determines that a candidate's progression is deemed unsatisfactory they shall take action IAW the training establishment's SOPs and shall be consistent with CFITES Vol. 6 - Conduct of Instruction Programmes.

TRAINING REVIEW BOARD

20. TRBs shall be convened and conducted IAW with NPTORD 4500-54 "Training Review Board Process".

21. In the event of prolonged trainee absence, TE Staff shall determine, on a case by case basis, the necessity to convene a TRB.

RESPONSIBILITY FOR ASSESSMENT

22. Assessment of trainees is to be done IAW CFITES Vol. 7 "Evaluation of Learners".

23. NFS staff provide quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NFS(P) Venture Division Command and Leadership Section.

COURSE REPORTS

24. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 "Course Reporting and Certificates" and amplified in NPTORD 4500-51 "Course Reports and Certifications".

25. Students will be graded complete or incomplete. The narrative will unpack the student's strengths and weaknesses based on PO/PC results, and will include specific remarking pertaining to the student's future potential for sea command.

26. Training limitations and/or incomplete training will be included in the course report.

CHAPTER 3 **- PERFORMANCE AND ENABLING OBJECTIVES**

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE NAVAL TRAINING DEVELOPMENT CENTRE (PACIFIC)

PO 001 – APPLY COMMAND AND CONTROL FOR WARSHIP COMMAND.

1. Performance Apply Command and Control for Warship Command.

2. Conditions:

   a. Given:

      (1) References as required

      (2) Heads of Department and other Subject Matter Experts.

   b. Denied: Nil;

   c. Environment: Any.

3. Standard: The officer shall apply command and control for warship command, at the ship level, by:

   a. applying regulations and doctrine of command authority;

   b. leading the ship; and

   c. solving command decision making problems, involving risk, ethical dilemmas and operations.

4. Training Time: Training time for this PO should not exceed 20% of the allocated training time – or 10 days of the 50 allocated.

5. Training Limitations: Nil

6. Summative Test Details: CEC 001, and PC 001 is run concurrently with PC 002D. The intent is to assess command risk appreciation and decision making under the evaluation of current major/minor warship COs. The examining officer, normally Cmdt NFS(P), must be satisfied that the officer is able to safely exercise auxiliary command, day/night, in all conditions.

PO 002 – APPLY PROFESSIONAL MARINER SKILLS FOR WARSHIP COMMAND

7. Performance: Apply professional mariner skills for warship command.

8. Conditions:

a. Given:

(1) References as required,

(2) A ship and its company.

b. Denied: Nil;

c. Environment: Any.

9. Standard: The officer shall apply professional mariner skills for warship command by:

a. applying International Collision Regulations and Canadian Modifications;

b. employing navigation policies and procedures;

c. applying good seamanship and ship-handling principles and practices, including weather and stability considerations; and

d. employing Bridge Resource Management (BRM) practices and procedures.

10. Training Time: Training time for this PO should not exceed 60% of the allocated training time – or 30 days of the 50 allocated.

11. Training Limitations: Nil

12. Summative Test Detail:

a. CEC – Collision Avoidance (NABS);

b. CEC – Shiphandling;

c. CEC – Command Board;

d. PC 002A - Collision Regulations;

e. PC 002B – Navigation;

f. PC 002C – Seamanship and Shiphandling; and

g. PC 002D – Practical examination of PO 002.

PO 003 – APPLY MARITIME LAW, USE OF FORCE, AND TACTICAL MISSION AND DEPLOYMENT PLANNING CONSIDERATIONS

13. Performance: Apply maritime law, use of force, and tactical mission and deployment planning considerations.

14. Conditions:

a. Given: References.

b. Denied: Nil;

c. Environment: Any.

15. Standard: The officer shall apply maritime law, use of force, and tactical mission and deployment planning considerations by:

a. directing tactical level mission planning;

b. applying national and international law to operational missions;

c. applying CAF doctrine to domestic operations;

d. directing ships damage control organization; and

e. directing ships company in the application of force.

16. Training Time: Training time for this PO should not exceed 10% of the allocated training time – or 5 days of the 50 allocated

17. Summative Test Details: Two (2) ECs; Maritime Law and Use of Force.

PO 004 – APPLY COMMAND-LEVEL MANAGEMENT TO SHIP'S RESOURCES

18. Performance: Apply command-level management to ship's resources.

19. Conditions:

a. Given:

(1) References;

(2) Heads of Department;

(3) Subject matter experts.

b. Denied: Nil;

c. Environment: Any.

20. Standard: The officer shall apply Command-level managerial processes and mechanisms applicable to commanding HMC Ships by:

a. overseeing force generation process as it applies to ships progressing to high readiness;

b. managing ship resources, including:

(1) finances;

(2) materiel; and

(3) food services.

c. overseeing personnel administration and discipline; and

d. overseeing engineering maintenance, configuration management, and change processes.

21. Training Time: Training time for this PO should not exceed 10% of the allocated training time – or 5 days of the 50 allocated

22. Training Limitations: Nil.

23. Summative Test Details: EC related to para 21 a-e.

PO 001 – Apply knowledge of Command and Control for Warship Command.

EO 001.01 Apply Regulations and Doctrine of Command Authority

1. Performance: The officer shall apply regulations and doctrine of command authority.

2. Conditions:

a. Given:

(1) References;

(2) Training resources;

(3) Training area;

(4) Case studies.

b. Denied: Nil;

c. Environment: classroom, NABS, ORCA.

3. Standard: The student shall apply regulations and doctrine of command authority at the ship level, by:

		Methods	Media	Environment	Class /Instructional Time	References
a	Explaining Command Authorities	Research/Lecture/GD			1	
b	Explaining National and International Command and Control Relationships	Reading assignment and guided discussion			1	
c	Explaining Command, Charge and Control from the Command perspective	Research and MSI Lecture			2	
d	Planning for Command Succession	In class Exercise			2	
Instruction time					6	

EC time	2*	
Total "in class" time	10	
Estimated Home Work time	6	

***Time for this EC not accounted for here.**

4. Formative Test Details.

   a. EC 001.01 – Draft Night Orders. The Officer shall be tasked to write Night orders for at least three different scenarios over the course of the classroom phase.

   b. CEC – Command Board.

5. Related Summative test Details: PC 001

EO 001.02 Lead the Ship

1. Performance: Lead the Ship.

2. Conditions:

   a. Given:

      (1) References;

      (2) Training resources;

      (3) Training area;

      (4) Case studies;

   b. Denied: Nil;

   c. Environment: classroom, NABS, ORCA.

3. Standard: The candidate shall, IAW references, lead HMC Ships by:

Teaching Points		Methods	Media	Environment	Time (in periods)	References
a.	Examining Distributed and Values-based Leadership	Reading Assignment and General Discussion			2	Leadership in the Canadian Forces Conceptual
b.	Assessing Organizational Leadership within the ship.	Lecture			1	SSOs
c.	Explaining Fighting Versus Functional					
d.	Explaining HODs responsibilities under the Fighting and Functional organizations					

e.	Describing A Fleet Commander's expectations of the Commanding Officer	Guest Speaker - Flag Officer	2	
f.	Describing the Duties of the Cox'n, and a Cox'n's expectation of the Commanding Officer	Guest Speaker – Fleet Coxn	1	
g.	Describing Leadership challenges at various readiness levels	Guest Speaker – Sea Training	1	
h.	Developing a Personal Command Philosophy	Course assignment – handed in prior to EC 001 (Practice Board)	1	
i.	Exemplifying Naval Leadership	Guest / Dinner	NA	
Instruction time			8	
EC time			0	
Total "in class" time			8	
Estimated Homework time			14	

4. Formative Test Details: EC 001.02 – Reflective Exercise: Write a Personal Command Philosophy.

5. Related Summative test Details: PC 001

6. Remarks: Guest speakers where practicable will coincide with lunch or other periods to allow for extended social interaction and discussion.

EO 001.03 Solve Command Decision-Making Problems Involving Risk, Ethical Dilemmas and Operations.

1. Performance: Solve command decision-making problems involving risk, ethical dilemmas and operations. .

2. Conditions:

a. Given:

(1) References;

(2) Training resources;

(3) command decision problems;

(4) Training area;

b. Denied: Nil;

c. Environment: classroom, NABS, ORCA.

3. Standard: The candidate solve command decision-making problems involving risk, ethical dilemmas and operations by:

Teaching Points		Methods	Media	Environment	Time (periods)	References
a.	Assessing Decision making techniques and problem analysis	Lecture			2	
b.	Analysing Risk and risk management– applying risk calculus to operational decisions	Lecture and discussion with case study			3	
c.	Applying the CF code of Ethics to ethical dilemmas	Read-ahead and case study			1	
d.	Assessing Case Studies in Command Decisions	Predominantly risk decisions – "Whiteboard Scenarios" see also PO 002			3*	

e.	Assessing Decision making in Command – a Senior officer's perspective	Senior Officer Presentation	1	
Instruction Time			10	
EC Time			8	
Total "in class" time			18	
Estimated Homework time			4	

*** Shared time with PO 002.**

4. Formative Test Details:

   a. EC 001.03 – The officer will be required to make and defend two decisions (one ethical, one risk) in a written exercise using a case study approach;

   b. CEC – Command Board.

5. Related Summative test Details: PC 001

PO 002 – Apply Professional Mariner Skills for Warship Command

EO 002.01 Apply International Collision Regulations and Canadian Modifications.

1. Performance: Apply International Collision Regulations and Canadian Modifications

2. Conditions:

   a. Given:

      (1) References;

      (2) Training resources;

      (3) Training area;

      (4) Case studies;

      (5) ORCA sea phase

   b. Denied: Nil;

   c. Environment: classroom, NABS, ORCA.

3. Standard: The officer shall Apply International Collision Regulations and Canadian Modifications, in a command role by:

Teaching Points		Methods	Media	Environment	Class/ Instructional Time	References
a.	Analysing COLREGs and explanatory texts and applying them to their own experiences	Instructor led discussion based on the completion of worksheets			6	Cockrotf, Farwell's and others
b.	Demonstrating knowledge of the text of Collision Regulations	Threshold knowledge COLREGs test and review			2	COLREGs

c.	Applying COLREGs during collision avoidance scenarios	Instructor led lecture on Collision avoidance and decision making supported by practical (simulated) collision avoidance scenarios	6	
d.	Assessing an application of the COLREGS to a case study	Case studies included in worksheets	2	
Instructional Time			16	
EC Time			1	
Total "in class" time (including simulator)			20	
Estimated Homework time			6	

4. Formative Test Details. EC #1 – 4. The student will be required to solve Command level COLREG problems presented in the NABS.

5. Related Summative Tests:

a. PC 002A – COLREG Examination;

b. EC 002 – Collision Avoidance Practical Simulation; and

c. PC 001 – ORCA sea phase.

EO 002.02 Employ Navigation Policies and Procedures

1. Performance: Employ navigation policies and procedures, to command navigation considerations, at the ship level.

2. Conditions:

a. Given:

(1) References;

(2) Training resources;

(3) Training area;

(4) Case studies;

(5) ORCA sea phase

b. Denied: Nil;

c. Environment: classroom, NABS, ORCA.

3. Standard: The officer shall employ navigation policies and procedures to command navigation considerations, at the ship level, by:

Teaching Points		Methods	Media	Environment	Time (Periods)	References
a.	Demonstrating familiarity with navigation policy, procedures and best practices across all three navigation regimes (pilotage, coastal and open ocean)	Discussion facilitated using worksheets or lectures			6	
b.	Demonstrating an understanding of the minimum level planning required to keep an ORCA class vessel safe and adhere to requirements of navigation policy	NPTT Simulator - Plan a short notice ORCA passage			3	

c.	Supervising and executing an ORCA class coastal navigation passage.	Supervise and execute a coastal navigation passage during ORCA class sea phase	1	
Instructional time			9	
EC Time			1	
Total "in class" time			10	
Estimated Homework time			6	

4. Formative Test Details: EC #1 – Plan and Execute an ORCA class coastal navigation passage in a simulated environment.

5. Related Summative Tests:

   a. PC 002B Navigation Exam;

   b. EC 002B Simulated Navigation passage;

   c. PC 001 ORCA sea phase; and

   d. PC 002D ORCA sea phase (OIC checklist).

EO 002.03 Apply Good Seamanship and Ship-handling Principles and Practices

1. Performance: Apply good seamanship and ship-handling principles and practices.

2. Conditions:

   a. Given:

      (1) References;

      (2) Training resources;

      (3) Training area;

      (4) Case studies;

   b. Denied: Nil;

   c. Environment: classroom, NABS, ORCA.

3. Standard: The candidate shall apply good seamanship and ship-handling principles and practices including weather and stability considerations by:

Teaching Points		Methods	Media	Environment	Time (periods)	References
a.	Answering a Threshold Knowledge exam	Written			2	
b.	Examining General Knowledge	Seamanship worksheet #1 (self-study assignment) and in class student presentation			2	
	Explaining RCN Seamanship Organization					
	Assessing Ship-handling Characteristics					
	Discussing use of Lines and Hawsers					

	Assessing Narrow and Confined Waters considerations			
	Explaining Shipboard organization			
c.	Explaining Formation Manoeuvring	Lecture and NABS Screenex (not an EC)	4	
	Explaining Tactical Signals			
	Explaining Formation control and direction as OCS			
d.	Explaining Berthing considerations	Seamanship Worksheets #2 through #4 (self-study assignment) and in class student presentation. Whiteboards – Case Studies requiring student analysis and classroom presentation Shiphandling Case Studies ECs in NABs and at Sea (time not reflected at right)	9	
	Planning an alongside or departure			
	Explaining a "standard" approach and departure – no set			
	Assessing On and Off-Jetty sets			
	Explaining Berthing using tugs			
	Explaining Berthing using an anchor			
	Explaining a Mediterranean mooring considerations			
e.	Explaining Anchoring considerations	Seamanship Worksheet #5 (self-study assignment) and in class student presentation.	1	
f.	Explaining Mooring considerations	Seamanship Worksheet #6 (self-study assignment) and in class student presentation.	1	

g.	Describing Replenishment considerations	Seamanship Worksheets #7 (self-study assignment) and in class student presentation. RAS Approaches in NABS (not an EC)	4	
h.	Explaining Towing considerations	Seamanship Worksheet #8 (self-study assignment) and in class student presentation. Whiteboards – Case Studies requiring student analysis and classroom presentation EC in NABS (time not reflected at right)	3	
i.	Explaining Weather considerations	Lecture	2	
j.	Explaining Rescue considerations	Lecture Whiteboard – Case Study requiring student analysis and classroom presentation	2	
10	Explaining Ice considerations	Lecture	1	
Instructional time			31	
EC Time			12 per student*	
Total "in class" time (Classroom and NABS)			43	
Estimated Homework time			20	

* Conducted in NABS individually for each student, these EC are run concurrent with other activities and so these 12 periods should not be accounted for as single activity time

4. Training Aids. The principle training aids involved in this EO are:

a. Seamanship and Ship-handling Worksheets: a series of worksheets intended to guide students through individual study, they contain a complete list of linked references and a list of questions to be answered. Students complete these on their own time and are required to be ready to present their results in scheduled class time;

b. Whiteboards are complete scenarios outlining a problem which require the student to conduct two activities - make a command decision as to whether they would (given an operational incentive) conduct the evolution proposed; and as a necessary precursor to that decision, determine how they would conduct it. Whiteboards therefore support at least two EOs: EO 001.03 (Make Command Decisions) and the EO relating to the skill at hand. In most cases this is EO 002.03

c. Ship-handling Case Studies are selected, short cases studies in ship-handling, assigned to make the students become familiar with the types of case studies contained in the references (Pronotes, Crenshaws naval Ship-handling, etc) and to increase exposure to ship-handling examples. Students are required to make a short 5 minute presentation on the assigned case study.

d. NABS Ship-handling Scenarios – as outlined in Annex H;

e. Live ship-handling practice in the ORCA Sea phase.

20. Formative Test Details. The following ECs are to be conducted individually in NABS (details and standards in Annex H):

(1) CEC – Two (2) Ship-handling scenarios in NABS; and

(2) CEC – Command Board.

21. Related Summative Test;

a. PC 002C Seamanship Examination - seamanship and ship-handling theory; and

b. PC 002D – ORCA Sea Phase. The ORCA OIC will examine practical application of Seamanship and Ship-handling.

22. Remarks. See Annex H for derivation and statement of NABS requirements.

EO 002.04 Employ Bridge Resource Management Practices and Procedures

1. Performance: Employ Bridge Resource Management practices and procedures.

2. Conditions:

a. Given:

(1) References;

(2) Training resources;

(3) Training area;

(4) Case studies;

b. Denied: Nil;

c. Environment: classroom, NABS, ORCA.

3. Standard: Employ Bridge Resource Management practices and procedures by:

	Teaching Points	Methods	Media	Environment	Time (Periods)	References
a.	Explaining the efficient use of bridge equipment:	Threshold Knowledge test and review followed by self study			2	
	ECPINS					
	Radar					
	GMDSS					
	Echo Sounder					
	Data Recording					
b.	Explaining Bridge Resource Management:	Research, Lecture and Guided Discussion			1	
	Basic Concepts					
	Error Chain Analysis					

c.	Applying BRM in Operations: Bridge Organisation and Functions Lines of Communications Training and Delegation Operations, evolutions and emergencies	Lecture & Guided Discussion	2	
Instructional time			5	
EC Time			2	
Total "in class" time			7	
Estimated Homework time			3	

4. Formative Test Details: CEC 002A, and 002C.

5. Related Summative Tests: PC 002D ORCA Sea Phase. ORCA OIC Certification. This certification will examine the practical application of BRM skills

PO 003 – Apply Maritime Law, Use of Force, and Tactical Mission and Deployment Planning Considerations

EO 003.01 Direct Tactical Level Mission Planning

1. Performance: Direct tactical level mission planning.

2. Conditions:,

   a. Given:

      (1) References;

      (2) Training resources;

      (3) Training area;

      (4) Case studies;

   b. Denied: Assistance;

   c. Environment: classroom, NABS, ORCA.

3. Standard: The student shall direct tactical level mission planning by:

Teaching Points		Methods	Media	Environment	Time (Periods)	References
a.	Applying tactical planning processes and the role of the captain	Lecture and classroom exercise			4	
b.	Initiating directions to ships team					
c.	Conducting a Commanding officer's personal estimate					
d.	Explaining CO's Role in Course of Action Development					
Instructional time					4	
EC Time					2	

Total “in class” time	4	
Estimated Homework time	0	

4. Formative Test Details: CEC – Command Board.

5. Related Summative Tests: PC 002D ORCA Sea Phase. ORCA OIC Certification. This certification will examine the practical application of tactical mission planning.

EO 003.02 Apply National and International Law with Respect to Maritime Operations.

1. Performance: Apply National and International law with respect to maritime operations.

2. Conditions:

a. Given:

(1) References;

(2) Training resources;

(3) Training area;

(4) Case studies;

b. Denied: Nil;

c. Environment: classroom, NABS, ORCA.

3. Standard: The candidate shall apply national and international law with respect to maritime operations by:

Teaching Points		Methods	Media	Environment	Time (Periods)	References
a.	Explaining Key Concepts in Maritime law	Self-Study			0	
b.	Explaining Legal Status of a Warship: rights, obligations, immunities	Lecture			1	
c.	Explaining Law of Armed Conflict in the Maritime domain	Self-Study			0	
d.	Explaining Piracy & Maritime Interdiction	Lecture			1	
e.	Explaining Maritime Law and the Arctic	Lecture			1	

f.	Explaining Commanding officers liabilities under Canadian Law pertaining to the Environment and safety.	Self-Study	0	
Instructional time			3	
EC Time			2*	
Total "in class" time			5	
Estimated Homework time			3	

4. <u>Formative Test Details:</u>

   a. EC 003.02 Maritime Law Worksheet – 2 periods.

EO 003.03 Apply CAF Doctrine to Domestic and International Operations.

1. Performance: Apply CAF doctrine to domestic and international operations.

2. Conditions:

a. Given:

(1) References;

(2) Training resources;

(3) Training area;

(4) Case studies;

b. Denied: Nil;

c. Environment: classroom, NABS, ORCA.

3. Standard: The officer shall apply CAF doctrine to domestic and international operations, by:

Teaching Points		Methods	Media	Environment	Time (Periods)	References
a.	Explaining Federal maritime jurisdictions in Canada:	Lecture and Self Study (worksheet)			1	
	Public Safety Canada and the Maritime emergency response plan (MERP)					
	RCMP					
	Transport Canada					
	DFO					
	CBSA					
	Coast Guard					

b.	Explaining the Conduct of Missions in support of OGD	Guest lectures / Guided Discussion	2	
	Explaining Canada Command authorities for domestic operations			
	OGD - Navy interoperability			
c.	Explaining Expeditionary Operations	Guest lectures / Guided Discussion	2	
	Explaining Command and Control under CEFCOM employment			
	Planning for deployed operations			
Instructional time			5	
EC Time			0	
Total "in class" time			5	
Estimated Homework time			2	

4. Formative Test Details: Nil.

EO 003.04 Direct the Damage Control Organization

1. Performance: Direct the Damage Control Organization

2. Conditions:

   a. Given:

      (1) References;

      (2) Training resources;

      (3) Training area;

      (4) Case studies;

   b. Denied: Nil;

   c. Environment: classroom, NABS, ORCA.

3. Standard: The candidate shall direct the damage control organization, by:

Teaching Points		Methods	Media	Environment	Time (periods)	References
a.	Explaining The Damage Control Organization	Lecture			2	
	Explaining Emergency Stations					
	Explaining Action Stations					
	Explaining HQ1					
	Explaining ERT					
	Explaining Casualty Clearing					
	Explaining DC Communications					
	Explaining Damage Control Prioritization					
b.	Explaining MSE/CSE considerations					
	Explaining Stability	Lecture			2	

	Explaining Power generation and Propulsion	Lecture	1	
	Explaining Combat systems	Lecture	1	
c.	Explaining the Response to Collisions, Groundings, and Fires in Peacetime	Lecture and Case studies	2	
Instructional time			8	
EC Time			0	
Total "in class" time			8	
Estimated Homework time			0	

4. Formative Test Details: Nil

5. Remarks. Candidates will have all platform backgrounds: major warship, minor warship and submarine. Classroom training should include as much non platform-specific instruction as practicable. There is no intent for this to evolve into a class-specific exercise.

EO 003.05 Direct Ships Company in the Application of Force

1. Performance: Direct Ships Company in the application of force.

2. Conditions:

   a. Given:

      (1) References;

      (2) Training resources;

      (3) Training area;

      (4) Case studies;

   b. Denied: Nil;

   c. Environment: Any.

3. Standard: The candidate shall, in accordance with references, direct ships company in the application of use of force knowledge, by:

Teaching Points		Methods	Media	Environment	Time (In periods)	References
a.	Explaining the key concepts from the command perspective	Self-Study - Use of Force Worksheet			0	
b.	Explaining Use of force in Domestic Operations	Lecture			1	
	Explaining Legal mandates					
	Explaining Use of Force					
c.	Explaining Use of Force in International and Combined Operations	Guest Lecturer – recently deployed			1	
d.	Explaining Command Management of the Fighting Organization	Lecture			1	

e.	Explaining Training, Controlling and directing individual use of force in Force Protection	Lecture – Sea Training	1	
Instructional time			6	
EC Time			2*	
Total "in class" time			8	
Estimated Homework time			2	

4. Formative Test Details:

a. EC 003.05 Use of Force Worksheet – a guided reference search and discussion.

PO 004 Apply Command-Level Management to Ship's Resources

EO 004.01 Oversee Own-ship's Force Generation to High Readiness

1. Performance: Oversee own ship's force generation to high readiness.

2. Conditions:

   a. Given:

      (1) References;

      (2) Training resources;

      (3) Training area;

      (4) Case studies;

   b. Denied: Nil;

   c. Environment: classroom, NABS, ORCA.

3. Standard: The candidate will oversee own ship's force generation to high readiness Force Generation, by:

Teaching Points		Methods	Media	Environment	Class/ Instructional Time	References
a.	Explaining the role of L1 Organizations:	Lecture (VTC) / Briefing Notes*			3	
	DGNSR					
	DGNFD					
	DGMEPM					
b.	Explaining the role of L2 Organizations					
	J3 – Operational focus	Guest Lectures / Briefing Notes*			3	
	J2 – Intelligence and service products					

	J5/J7 – Operations planning			
	J6 – Communication and information systems			
c.	Explaining the RCN Force Generation Process including			
	- Unit Organization and manning	Guest Lecture J1 / Briefing Note*	2	
	REMAR			
	Establishment			
	Manning Priorities			
	ER and SR manning challenges			
	Critical manning			
	- Readiness and Sustainment Programme	Guest Lecture J3 / Briefing Note*	2	
	Readiness levels			
	Responsibilities			
	Scheduling			
	Reporting			
	Workups Processes			
	Explaining Technical support in Force Generation	Guest Lecture – J4 FMF	2	
Instructional time			12	
EC Time			2	
Total "in class" time			14	
Estimated Homework time				

*When scheduling constraints to not permit time for lectures the Briefing Notes will suffice.

4. Formative Test Details. Nil.

EO 004.02 Manage ship's resources

54. Performance: Manage Ships Resources.

55. Conditions:

a. Given:

(1) References;

(2) Training resources;

(3) Training area;

(4) Case studies;

b. Denied: Nil;

c. Environment: classroom, NABS, ORCA.

56. Standard: The candidate shall manage ship resources, in HMC Ships, including finances, material and food services, by:

Teaching Points		Methods	Media	Environment	Time (Periods)	References
a.	Explaining Financial Oversight:	Lecture J4 (Log)			2	
	Public					
	Non Public Funds					
	Funds, Grants and Charities					
	Canteen and Duty Free (Tobacco and Alcohol)					
	Hospitality					
	Audits					
	Attestation					
	Contracting					
	Food services					
	Capability Planning and Budgets	Lecture J8			1	
b.	Explaining the control of: Materiel	Lecture F4			1	

	Fitted equipment list (FEL), Supply Customer Account (SCA), Operational Equipment			
	Fuel			
	Ammunition			
	IORs			
	Operational Deficiency reporting (OPDEF)			
c.	Explaining Safety and Environment considerations:	Lecture Formation RiskMO / Briefing Note*	1	
	Fleet and Formation Safety organizations			
	Environmental and Safety Regulations			
Instructional time			5	
EC Time				
Total "in class" time			5	
Estimated Homework time				

* When scheduling constraints to not permit time for lectures the Briefing Notes will suffice.

57. Formative Test Details. EC 004.02 Manage Ship's Resources Worksheet – 2 periods.

EO 004.03 Oversee Personnel Administration and Discipline

58. Performance: Oversee Personnel Administration and Discipline.

59. Conditions:

a. Given:

(1) References;

(2) Training resources;

(3) Training area;

(4) Case studies;

b. Denied: Nil;

c. Environment: classroom, NABS, ORCA.

60. Standard: The candidate shall oversee personnel administration and discipline in accordance with CAF regulations, policies and practices in HMC Ships, by:

Teaching Points		Methods	Media	Environment	Time (Periods)	Refs
a.	Explaining the Management of the Functional Organization	Lecture / Briefing notes			2	
	HODs Responsibilities under the Functional Organization					
	Key Committees and organizations					
	Importance Ships books					
	Oversight of Divisional Records					
	Personnel Records					
	- Explain CO's role in security management:	Lecture (DPM Secur) / Briefing notes*			1	
	IT security					
	Physical security					
b.	Explaining Personnel policies:				1	

	Harassment and conflict resolution	Lecture (DRC)/ Briefing notes*		
	Explain the resolution procedures			
	Family Support	Lecture (MFRC) / Briefing notes*	1	
c.	Explaining Remedial Measures and Administrative Reviews	Lecture(J02 SSO Admin)	3	
d.	Explaining Medical Administration	Lecture (Formation Health Services Officer)	2	
	CO's role in the medical process			
	On-board Capabilities and limitations			
	Health Services oversight			
	CF98, DND 663, MELs			
	Deployed ops medical considerations			
e.	Explaining Discipline	Presiding Officers Course	14	
Instructional time			21	
EC Time			3	PSO Test
Total "in class" time			24	
Estimated Homework time				

* When scheduling constraints to not permit time for lectures the Briefing Notes will suffice.

61. Formative Test Details. Nil.

EO 004.04 Oversee Maintenance and Engineering Activities

1. Performance: Oversee maintenance and engineering activities.

2. Conditions:

   a. Given:

      (1) References;

      (2) Training resources;

      (3) Training area;

      (4) Case studies;

   b. Denied: Nil;

   c. Environment: classroom, NABS, ORCA.

3. Standard: The officer shall oversee maintenance, and engineering activities in HMC Ships, by:

Teaching Points		Methods	Media	Environment	Time (Periods)	References
a.	Explaining RCN Engineering Governance	Lecture (VTC) DGMEPM			2	
b.	Explaining Configuration management	Lecture (J4 Eng)			2	
	Mission fits					
	EC process					
c.	Planning maintenance	Lecture (F4 Eng)			2	
	Explaining Shipboard planned maintenance organization					
	Work periods planning					
	Explaining CM and PM					
	Explaining contracted maintenance					
	Explain condition based maintenance					

Instructional time	6	
EC Time		
Total “in class” time	6	
Estimated Homework time		

4. Formative Test Details. PC 004 – Scenario based problems requiring identification of problem, initial reaction/direction and knowledge of support agency (ies).

ANNEX A – MAIN REFERENCES

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PO 003 Apply Maritime Law, Use of Force, and Tactical Mission/Deployment Planning Considerations
DC Manual Volume 1 Annex Y
Damage Control Instructions
Ship Standing Orders
Development and Evaluation of an Intuitive OPP - DRDC CR2006-076
CF Operations Planning Process
Criminal Code of Canada
Convention on the High Seas (1958)
International Convention for the Safety of Life at Sea (SOLAS) (1974)
UN Convention on the Law of the Sea (UNCLOS) (1982)
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PO 004 Apply Command Level Management Theory for Warship Command
National Defence Security Policy (NDSP)
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DAOD 5012-0 / HA Prevention & Resolution Guidelines
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Military Family Resource Centre www.familyforce.ca
DAOD 7002-1 Boards of Inquiry, Summary Investigations

DAOD 7002-3 Investigative Matters and References
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DC Pol Contract Advisory Bulletin # 6
CF Food Services Manual A-85-269-001/FP-001
The Financial Administration Act
Naval Engineering Manual (NEM)
Canadian Environmental Protection Act: Federal Halocarbon Regulations
Environmental Directive 4003-5 Halocarbon Management

ANNEX B – TASK LIST

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ANNEX C – QSP BOARD RECORD OF PROCEEDINGS

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ANNEX D – QSP WRITING BOARD

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ANNEX E – ON JOB PERFORMANCE REQUIREMENTS INSTRUCTIONS

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ANNEX F – ACRONYMS

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ANNEX G - SUGGESTED SEQUENCING AND TIMINGS

Scheduling constraints/imperatives:

1. PO 001 Command a Warship should be concentrated in the first third of the course.

2. The course should be scheduled to avoid sole subject concentrations in one period of time. A graduated approach of overlapping instruction has been found to work most effectively in terms of student interest and retention. For example, collision regulations are taught daily for short periods (student led quizzes, short tests). The seamanship and shiphandling material should be distributed throughout the course, and interspersed with lectures relating to PO 003 and 004.

3. NABS training should be half days where possible, allowing for lectures in the forenoon followed by practical training in the afternoon.

3. PC 002A (Collision Regulations), PC 002B (Navigation) and PC 002C (Seamanship) shall be completed at least 3 weeks before the end of the course to facilitate re-write opportunities.

4. The ORCA Sea Phase should be conducted in the last two (2) weeks of the course – representing as it does the culminating evaluation phase of the course.

ANNEX H - TRAINING RESOURCES

1. In accordance with Chapter 1 paragraph 5 add any extra tables or paragraphs used for Resources in this annex. (**Not** to be deleted.)

Navigation and Bridge Simulator Requirements

2. The accompanying table sets out NABS training requirements derived from the EC as laid out in Chapter 3. Note that where the ECs involve individual students solving of a shiphandling problem or keeping a scenario secret (ie: not sharing collision avoidance scenarios in order to preserve the integrity of the EC).

3. The NABS bridges will require manning support (OOW, Helmsman) from other than CDC students and support staff through CFTPO framework, in order to also, preserve the integrity of the EC and learning process. Those serials are indicated below with an asterisk (*).

	Title	**Objective**	**Desired Student Time**	**Actual NABS Bridge Hours (16 students)**	**Total Elapsed NABS Time**	**Remarks**
1	Rules Practical Intro	1.Exposure to Appreciate/ Plan/Execute 2. practical examination of May/Shall in Rule 15	0.25 Hr*	4 (Assuming 2 Bridges)	4	*Plenary session. Discussion of Rule 15 and Cockcrofts et al commentary on the subject followed by exercise in applying the rule as Stand-on and Give way vessel.
2	Standard Approach	To calibrate the student to basic jetty approach conditions	1 Hr	16	20	No Set 'Infinite' Jetty Imposed Speed regime:

						Major: no less than 5 knots to 2 Cables Minor: no less that 4 knots to 2 Cables For Major warships: Flare to be conducted with rudder only, matched engines until checking the flare Minor Warships flare guidelines TBP 6. Student to remain at Centreline until throwing for No. 1
3	High Speed Approach	1.To develop confidence in stopping power of ship 2. Practice use of shadow zone in berthing	0.5 Hr	16	36	1. No Set 2. Ship ahead 3. Speed Regime: Major/Minor - NLT 10 Knots to 300 yards.
4.	Tow Approaches	To develop confidence in compensating for set. To develop skill in this manoeuvre.	0.5 Hr	8	44	1. Similar ship, similar drift 2. Wind NLT 25 knots 3. Run to start with own ship 1 nm back on the desired approach radial
5	Line Handling	1. To develop skill in handling lines alongside	0.75 Hr	12 Hrs	62	1. 20 Knot off jetty wind, no tugs 2. Ship to start 50 yards of jetty, 15 degrees into lie, No. 1 across.

						3. Student to practice twisting on one to bring ship alongside 4. Repeat scenario with No1 and 2 Across.
6 *	Rules A	Appreciate/ Plan/ Execute	0.5 Hr	8	70	Suggest Eastern San Juan two crossing tugs and tows In formation
7	Off Jetty Approach	1.Develop ability to respond to an off jetty set, with and without tugs 2. Value of Transits 3. Compensating for set 3. Instructing a tug	1.5 Hr	24 Hrs	94	1. 20 knot off jetty wind 2. straight in approach 3. At least two approaches without tug 4. At least one with tug 5. Includes line handling 6. Time permitting follow-on run with Quartering wind
8.	On Jetty Approach	Reading set by use of transits and instruments compensating on the approach altering onto an approach track	1 Hr	16 Hrs	110	20 knot on jetty at least 2 with straight-in approach to start with long approach (1000 yards) At least one with 90 degree a/c onto the approach (Halifax NC?)

9 *	Rules #B	Appreciate/ Plan/ Execute	0.5 Hr	8	118	Canso, at Night In formation
10.	Screene x	Allied Rules Joining from ahead RAS Approaches Plane Guard	2.0*	8 Hrs	126	* Plenary Session A 2.0 hours serial in 4 bridges concurrently Night Time AOR, CVN, at least 4 destroyers Scenario to exercise joining from ahead, Plane Guard, Moving to Standby and lifeguard station, RAS Approaches.
11	Shipha ndling with an Anchor	Develop confidence in use of an anchor to proceed alongside	0.5	8 Hrs	134	1. Students to pre-plan a starboard side to approach on Rocky Point Ammo Jetty using wind 090@25 knots. Plan to incorporate use of an anchor. This serial done in "plenary" Once the "staff' solution to this problem is fully identified, time allowing move on to a wind 000@25 knots, using anchor to set down onto the berth :
12 *	Shipha ndling proble m #1	Plan/Execut e	1 Hr	16 Hr	150	St Johns Harbour Move – No tugs Minors??? TBP
13 *	Shipha ndling	Plan/Execut e	0.75 Hr	12 Hr	162	Majors: A1 Extension with an off jetty wind,

	problem #2					ship at A1. One tug assist. Minors; Canada place nest
14 *	Shiphandling problem #3	Plan/Execute	0.75 Hr	12.0 Hr	174	Major: San Francisco Minor ; San Francisco
15 *	Rules #C	Appreciate/ Plan/ Execute	0.5 Hr	8	182	Point Atkinson
16 *	Practice Boards		2 Hrs**	32	214	** Note: 2 hours boards each but managing sched to conduct 4 per day
						2 ORCA (for multi-ship activity), 2 Current or past XO/CO as mentor evaluators, ORCA crews to facilitate overnight sailing

ORCA Requirements

Title	Days	Description of Requirements
ORCA – Sea Phase	10	2 ORCA (for multi-ship activity); 2 Current CO as OIC and mentor evaluators; and ORCA crews to facilitate overnight sailing.

ANNEX I - COSTING INFORMATION

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ANNEX J - COURSE HISTORY

1. The CDC course was first conducted in April 2010, after two QSP drafting boards were held in the fall of 2009, and CMS released MARGEN 048-09 ordered the implementation of the course and the phasing out of the extant Command Part I process (the Command Examinations).

2. The decision to go to a course for Command Part I Certification vice a self study and examination process was based on:

a. a realization that the Command Examinations were of dubious value and applicability in confirming command level professional knowledge;

b. concern that insufficient MARS officers were completing the Part I process, and

c. a desire to make more effective us of the 10 Months on the Advanced Training List (ATL) that the ORO course represented.

3. The decision was made to reduce the length of the ORO course in length and in the time saved, to insert a 10 week Command Development Course which would take the place of the Command Examination process, and award a Command Part I Certification as a result of successfully completing that course.

4. The intent was that officers selected for the ORO course would continue to spend 10 months on the ATL, but would proceed directly from the shortened ORO course to the CDC, and would join their ships as OROs on completion of the CDC. It was also intended that the CDC would cater to Submariners and Reserve Officers, who would attend the course at a similar point in their career progression to the OROs of the Regular Surface Fleet.

5. The first CDC was run in April through June 2010. The CDC QSP (November 2011) was revised after 4 courses had been conducted, with minor amendments throughout. Feedback from CDC students had been universally positive, but minor amendments were suggested, and the instructors implemented significant changes between the first and the fourth course.

6. First Revision to the CDC QSP (November 2011) included:

a. the placement more emphasis on Seamanship and Shiphandling, including a series of seamanship self-study assignments and a closed-book pass/fail PC (PC 002C);

b. The development of a rigorous programme of individual practice in NABS, following a graduated approach of compartmented shiphandling problems through to advanced planning and execution problems; and

c. Other changes are as summarized in the changes table to the End of Course report from CDC 1101 (ended July 2011), appended here: (items with a strike through line are those changes which had already been implemented following previous courses).

7. The most significant change in 2011 was the insertion into the CDC QS of the ORCA Sea-phase, and the adoption of a pass/fail element to that sea phase (see ANNEX L for full description). Previously, the ORCA Sea Phase, of two weeks duration, had been a part of the 10 week course, but it was not formally part of the QSP and there was no course implication to not achieving an ORCA OIC certification.

8. The ORCA Seaphase had originally been intended to serve two purposes: first, a need in the fleet for more ORCA certified officers (not a CDC requirement), and second, a valuable opportunity for the students to practice shiphandling. During the first four courses, however, the instructors observed that in addition to the expected benefits of ship driving, the officers were getting a much needed refresher on basic bridge practices (ECPINS, Radar Tuning, Ship to ship communications, etc) which was serving a real training/experience requirement. In addition, the sea phase offered an opportunity to build a programme which developed command decision making skills in an applied, practical fashion through inserting risk decision problems into the programme – challenges to the student captain which were not part of the ORCA OIC certification mandate.

9. The decision was therefore made to include the ORCA Seaphase as part of the QSP, with two specific goals:

10. EO 001B: Developmental training and evaluation in command decision making;

4. PC 002C: The ORCA OIC Certification process, under the direction of the Commander Coastal Forces was to be used for two purposes: to qualify the officer as an ORCA OIC, and to serve as a practical examination for PO 002.

Second Revision to the QSP (Jan 2014)

5. The second revision to the QSP was put in place for CDC 1401, commencing in January of 2014. Key changes were:

a. Introduction of a written pass/fail examination in navigation theory new PC 002B. Seamanship Exam to be re-numbered PC 002C.

b. Formalization of amendments to the seaphase, included:

(1) A requirement for current ship COs to act as Orca OIC and Mentor/evaluators; and

(2) EC 001B (see first revision) became a PC 001, formally stating the requirement to satisfy Commander NOTC that the officer was sufficiently versed in command decision making and risk management to operate an ORCA day/night. The Sea phase now consists of two PC – PC 001 and PC 002D.

Third Revision to the QSP (Dec 2020)

15. The third revision to the QSP was completed in the fall of 2020, and is being put in place for CDC 21-01, commencing January of 2021. Key changes were:

a. Condensing a four (4) chapter QSP, into a three (3) chapter QSP;

b. Alignment of POs/EOs, and Annex K AAP;

c. Assessment details in Chapter 2, and in Annex K have been updated to reflect the removal of the Command Board as a formal qualification process, based on the new Sea Command Potential Assessment (SCPA) process; and

16. Cumulative assessments have moved from knowledge to skill (practical) assessments.

ANNEX K – ASSESSMENT PLAN

Part 1 - Assessment Overview

The scope of the assessment represents portions of the EOs.
This assessment plan shall be read in conjunction with the Annex K, Part 2 AAP and Annex G, Suggested Sequencing and Timings.

ASSESSMENT OVERVIEW							
Scope	**Purpose of Check**	**Method Used**	**When and How Conducted**	**Resources**	**Time**	**Limitations**	**Impact & Remedial Instruction**
PC 001 – ORCA SEAPHASE PO 001	To assess the officer's competence in exercising the decision making, risk appreciation and management commensurate with commanding an ORCA in typical multi-ship day/night operations	Practical at Sea	See Annex K Part 2 and Annex G Part 2.	2 ORCA Class PCT and crew. 2 current or recent warship CO as assessing officers	16 students nominally, each acting as Captain for one day; 8 days Fuelling and Storing, 1 day Familiarization at sea, 1 day Total 10 days.		Failure results in TRB iaw NPTORD 4500-54
PC 002D (ORCA OIC) PO 002	To assess the officer's platform proficiency and competence in accordance with the standard for ORCA OIC Certification.	Practical at Sea			16 students nominally, each acting as Captain for one day; 8 days Fuelling and Storing, 1 day Familiarization at sea, 1 day Total 10 days.		Failure results in TRB iaw NPTORD 4500-54
CEC 001 **Scope:** **All POs**	To evaluate the student's ability to make command decisions in the field of all Performance Objectives	Oral Board	Week preceding the PC 001 PC002D (Seaphase).	Board consisting of Cmdt NFS(P), another current or former ships CO, 1 instructor	30 minutes prep time, 1 Hour execution	1 hour oral examination to exercise and evaluate performance in all POs with emphasis on 001,002,003.	Formative feedback from Board members, Performance reflected in Course report. This constitutes one of three (1/3) CECs.
PC 002A **Scope:** **PO 002**	To confirm knowledge of COLREGS	Written examination	In class, invigilated	None.	Total Time 90 min	Closed book	Exam Review with instructor Re-attempt Unsatisfactory result on PC will result in TRB IAW NPTORD 4500-54
CEC 002.01 **Scope:** **EO 001.03** **EO 002.01**	To confirm the student's ability to apply collision avoidance and decision making skills in a practical environment	NABS	Conducted throughout the course as four different scenarios that each student challenges without previous exposure to the scenario Collision avoidance scenarios A,B,C,D	Any simulated vessel and Bridge. Helmsman and OOW or Navigating Officer.	Total Time 120 min/student Each scenario requires two bridge days		Formative Feedback This constitutes one of three (1/3) CECs.
PC 002B **Scope:** **PO 002**	To Confirm student's knowledge of navigation policy and procedures	Written Exam	Invigilated; in class	Classroom	Total time 90 min	Closed book	Review Re-write 2 consecutive failures results in a TRB IAW NPTORD4500-54

EC 002.02 **Scope:** **EO 002.02** **PC 002D**	To confirm student's familiarity with ECPINS, SDM and bridge navigation equipment related failures. Preparation for ORCA sea phase.	Simulated navigation passage in NABS	Conducted as 2 simulated navigation passages in ORCA class vessels Students assigned scenario Present passage plan to instructor Plan and execute passage over 60min	NABS	Total time 3 bridge days		Peer and Instructor Formative Feedback A critical component of ORCA sea phase checklist (PC 002D). Opportunity to continue development during ORCA sea phase and be assessed.
PC 002C **Scope:** **PO 002**	To Confirm student's knowledge of ship handling and seamanship principles, policy, procedure, and best practices.	Written exam	Invigilated; in class	Classroom	120 mins	Closed book	Review Re-attempt 2nd unsuccessful attempt results in TRB IAW NPTORD 4500-54
CEC 002C	To evaluate the student's skills in shiphandling	Simulated shiphandling scenarios in NABS	NABS	NABS	90 mins	Nil	Formative Feedback This constitutes one of three (1/3) CECs.
EC 003.01 **Maritime Law**	To develop the students' knowledge of Maritime Law and principal references	Worksheet – open book quiz followed by classroom review and discussion	Classroom and Own time	1 Legal Officer	3 hours	Nil	Formative feedback by instructors and Legal Officer
EC 003.02 **Use of Force**	To develop the student's command level knowledge of the CF governance n Use of Force	Worksheet – open book quiz Followed by classroom review and discussion	Classroom Own time	1 Legal Officer	3 Hours	Nil	Formative feedback by instructors and Legal Officer

Part 2 - Activity and Assessment Plan (AAP)

PC 001 – ORCA Sea Phase

Overview	Objective(s)	Assessment Details	Lessons Identified
Media: Ship at sea **Method**: Practical **Time:** 10 Sea Days **Environment**: At Sea	• To assess the officer's competence in exercising the decision making, risk appreciation and management commensurate with commanding an ORCA in typical multi-ship day/night operations The assessing officer will evaluate the students' command appreciation as it relates to risk management, decision making and problem solving during anti-collision, navigation, shiphandling and other operational activities. The evaluator will also assess the officers' ability to participate in and lead a team effectively under conditions of stress and fatigue.	Assessment will be done by the Assessing officer over the duration of the 10 day sea phase. In addition, each student will spend a period of 24 hours as "Captain" for a dedicated assessment period. **<u>Before the assessment:</u>** The Student will be required to plan the day; **<u>During the assessment:</u>** This is a subjective assessment of the officer's competence in exercising decision making, risk appreciation and management commensurate with the commanding of an ORCA class vessel in typical multi-ship day/night operations. The Student will spend up to 24 hours as the Captain and will be assessed on the quality of their decision making and risk management as it pertains to the following common activities. It should be noted that students will be evaluated on their decisions, risk assessment and leadership throughout their time at sea, not just when acting as OIC. When considering the student's appreciation of risk it is important to remember that the mission is to maximize training opportunities in the time allotted. Students should be looking for ways to achieve the mission and should only cancel serials or activities in the unlikely scenario that a creative solution cannot be found. **<u>Assessment feedback:</u>** 1. The Assessing Officer will provide written feedback and assessment.	**To be developed** - Run sheet - Assessment Checklist
	Impact and Limitations	**Resources**	
	Impact: **Limitations / Remarks**:	**For student:** **Two ORCA PCTs and crew.** **- Two Assessor/Mentors who will act as OICs and should be current or recent warship COs.** **For instructor:** - Run sheet - Assessment Checklist	

CEC 001 – Oral Board

Overview	Objective(s)	Assessment Details		Lessons Identified
Media: none **Method**: Oral Examination **Time:** 30 Min Preparation 1 Hour Execution **Environment**: A Board Room	To evaluate the student's ability to make good command decisions in the field of all Performance Objectives	**Before the assessment:** The Student will be provided with a basic scenario and sufficient time to prepare **During the assessment:** The Student will be presented with tabletop scenarios relating to any of the Performance objectives of the Course, but with emphasis on PO 001 and PO 002. **Assessment feedback:** The Assessing Officer will provide written feedback and assessment.		
				To be developed
	Impact and Limitations	**Resources**		
	Impact: This EC will form part of the Officers Evaluation and will be reflected in the course report. **Limitations / Remarks**:	**For student:** Provided as necessary	Board consisting of Cmdt NFS(P), at least one other another current or former ships CO, and 1 senior instructor as member and Board Secretary	

EO 001.01 – Apply Command and Control Theory for Warship Command

Overview	Objective(s)	Activity Details	Lessons Identified
Media: none **Method**: Lecture and Case Study **Time**: 40 hours **Environment**: Classroom	The officer shall apply: a. detailed knowledge of the regulations and doctrine pertaining to command authority at the ship level; b. detailed knowledge of Command and Control doctrine in the CF and RCN c. detailed knowledge of CF and RCN theory and models for leadership at the ship command level d. basic competency in solving command decision making problems involving risk, ethical dilemmas, and operations	Activity# 1 – Study CF and RCN doctrine on Command and Control - 2 hours Activity #2 – Study theory and best practices of Decision making – 6 hours Activity #3 – Articulate a Personal Command Philosophy – 1 hour plus own time Activity #4 – Draft Night Orders – 4 hours classroom plus own time preparation Activity #5 – Conduct Command Scenario Decision Exercises The officer will be required to make general command decisions using real or fictitious case studies and scenarios – 16 hours Activity #6 - Conduct Risk Decision Exercises - 4 hours. The officer will be required to make and defend two decisions (one ethical, one risk) using a case study approach; Activity 7 – Conduct and Ethical Dilemma Decision Exercise 4 hours The officer will be required to make and defend a decision using a scenario or case study approach; Activity# 8 - Conduct Case Studies in Leadership Respect and Honour studies 3 hours	**To be developed**

Why	Resources	
Knowledge and practice of the objectives are vital to successful warship command.	**For student:** Key References Scenarios for Study and Exercise Visiting guest speakers appropriate to Case Studies.	**For instructor:**

PC 002A – Confirm knowledge of COLREGs

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Assessment Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="3">Media:
none

Method:
Written examination

Time: 90 min

Environment:
classroom</td><td>The goal of this PC is to assess the student's knowledge of COLREGs</td><td colspan="2"><u>Before the assessment:</u>
1. Student's will conduct a practice examination
2. Students will execute a series of worksheets designed to explore the rules. Discussion time allocated for the review of each worksheet.

<u>Assessment feedback:</u>
2. Formative review provided after the practice examination
3. Formative review provided after examination</td><td></td></tr>
<tr><td></td><td></td><td></td><td>To be developed</td></tr>
<tr><td>Impact and Limitations</td><td colspan="2">Resources</td><td rowspan="2"></td></tr>
<tr><td>Impact: Students must achieve 85%

Limitations / Remarks:
- No access to resources during the exam</td><td>For student:
- Pencil and pen
- Exam may be completed electronically if MS Forms is used in which case a DWAN laptop may be required.</td><td>For instructor:
- Exam
- Exam answer sheets
- If completed as a MS Form, link will need to be enabled for the period of the exam</td></tr>
</table>

EO 002.01 – Apply Collision Regulations

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Combination of student and instructor led depending on the activity **Media**: Classroom with projector NABS NABS theatre Sea phase **Method**: **Time**: - 6 x 1 hr period for discussion of worksheets - 1 bridge simulator day for demonstration and discussion - 8 bridge simulator days for individual collision avoidance scenarios - 1 hr/student for case studies - 10 day ORCA seaphase **Environment**: Classroom, NABs and at sea	1. Develop deep knowledge IRPCS and CANMODs 2. Translate decision making theory into the practical environment 3. Apply COLREGs in a practical environment <u>Teaching points:</u> 1. COLREGs 2. Decision making best practices for collision avoidance	<u>Activity #1 – COLREGS worksheets</u> Complete worksheets and discuss in class <u>Activity #2 – Collision Avoidance demonstration</u> Instructor to provide decision making and best practices during collision avoidance lecture Lecture followed by 4 simulated, practical collision avoidance scenarios, executed by students and reviewed in NABS theatre to highlight teaching points <u>Activity #3 – Individual collision avoidance scenarios</u> Students are presented with individual scenarios in NABs and are asked to resolve them using decision making techniques and the application of best practices. NABS theatre available for those who have already completed runs to observe <u>Activity #4 – ORCA sea phase</u> Students apply decision making and best practices at sea and are assessed in the form of PC 001A and 002D	
	Why	**Resources**	**To be developed**
	- Understanding COLREGs is fundamental to collision avoidance - Decision making in a high stress, time sensitive environment can be improved using best practices	**For student:** a. Access to reference library b. DWAN laptop **For instructor:** <u>NABS:</u> 1 operator/cubicle 1 helmsman/cubicle NABS theatre Classroom with DWAN and projector ORCA class vessels for sea phase	

CEC 002.01 – Apply Collision Regulations

Overview	Objective(s)	Assessment Detail		Lessons Identified
Media: NABS NABS theatre **Method**: **Time**: 4 Bridge days in NABS **Environment**: NABS	Assess students' ability to apply COLREGS, BRM and decision-making best practices to resolve complex vessel interaction scenarios	See Collision Avoidance Assessment Sheet -This assessment mimics the collision avoidance assessment conducted during Command Boards prior to 2020 -The scenarios are resident in NABS and are scenarios that have been used during past command board. -There are four scenarios/student, each presented individually -Prior to scenario start, the student receives a briefing sheet that describes the scenario -Each assessment takes approximately 15 min -The assessment is cumulative over the four scenarios and an assessment form is provided for each scenario -students who have completed their individual run can watch following students' runs from the NABS theatre -Assessment is conducted by one of the two senior Naval Instructors		
				To be developed
	Impact and Limitations	**Resources**		
		For student: Nil	**For instructor:** <u>NABS:</u> 1 operator/cubicle 1 helmsman/cubicle 1 OOW/Navigator NABS theatre	

PC 002B – Confirm knowledge of Navigation theory, policy and regulations

Overview	Objective(s)	Assessment Details		Lessons Identified
Media: none **Method**: Written Exam **Time**: 90 min	The goal of this PC is to assess the student's knowledge of navigation theory, policy and regulations.	**<u>Before the assessment:</u>** Students will execute a series of worksheets designed to explore the navigation theory, policy and regulations **<u>Assessment feedback:</u>** Formative Review		-
				To be developed
	Impact and Limitations	**Resources**		
	Impact: Students must achieve 85 % to pass **Limitations / Remarks**: - No access to resources during the exam	**For student:** - Pencil and pen - Exam may be completed electronically if MS Forms is used in which case a DWAN laptop will be required.	**For instructor:** - Exam - Exam answer sheets - If completed as an MS form, link will need to be enabled for the period of the exam	

EO 002.02 Direct Navigation

Overview	Objective(s)	Activity Details		Lessons Identified
Strategy: Combination of student and instructor led depending on the activity **Media**: Classroom with projector NABS NABS theatre Sea phase **Method**: EDIP **Time**: 3x2hr blocks for discussion of worksheets 4 Bridge Days in NABS for the practical application of planning and execution of pilotage **Environment**: Classroom NABS ORCA sea phase	- Develop deep knowledge of navigation policy, procedures and practices across all three navigation regimes - Apply knowledge of planning and execution of pilotage to keep a simulated ORCA class vessel safe. - Apply knowledge to keep an ORCA class vessel safe at sea.	<u>Activity# 1: Worksheets (3 x 2hrs)</u> Complete worksheets and discuss in class <u>Activity# 2: ORCA passages</u> Student is given an snap passage in the Gulf Islands and asked to set up navigation equipment, plan a short passage then navigate the ORCA.. Student will encounter contacts and restricted visibility during the passage First 30 min of allocated time is for planning and set up, the second 30min is for execution and de-brief. <u>Activity #3: Pilotage passages during ORCA sea phase</u> Students will plan and conduct pilotage and coastal navigation passages. <u>Activity# 4: Case Studies:</u> Students analyse case studies and present them to the class		c.
	Why	**Resources**		**To be developed**
	-Understanding of navigation policy, procedures and best practices is fundamental to safe navigation and the exercise of Commanding Officers authority -Exposure to practical navigation prepares students for ORCA sea phase	**For student:** NABs cubicle for 30min of planning and set-up prior to run commencement	**For instructor:** Assessment checklist FNI or someone capable of assessing a navigation passage	

PC 002C – Seamanship and Shiphandling Examination

Overview	Objective(s)	Assessment Details	Lessons Identified
Media: **Method**: Written Examination, closed Book **Time**: 2 hours	To assess the officers professional knowledge in Seamanship and Shiphanding.	Assessment will include the following topics; 1. General knowledge of the Seamanship organization in the RCN 2. Ship performance characteristics 3. Line handling 4. Berthing and Departures 5. Boats 6. Replenishment 7. Towing 8. Anchors and their use 9. Buoy work 10. Ship Stability 11. Weather	-
			To be developed
	Impact and Limitations	**Resources**	
	Impact: Students must achieve 85 % to pass **Limitations / Remarks**: No access to resources during the exam	**For student:** - Pencil and pen - Exam may be completed electronically if MS Forms is used in which case a DWAN laptop will be required. **For instructor:** - Exam - Exam answer sheets If completed as an MS form, link will need to be enabled for the period of the exam	

EO 002C Apply good seamanship and ship handling principles and practices

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Combination of student and instructor led depending on the activity **Media**: Classroom with projector NABS NABS theatre Sea phase **Method**: **Time**: 15 hours classroom time split amongst the 8 worksheets and supporting Instructor lectures 20 Bridge Days in NABS for practical application of shiphandling and seamanship skills pilotage **Environment**: Classroom NABS ORCA sea phase	- Develop deep knowledge of Seamanship and Shiphandling policy, procedures and practices - Apply knowledge of planning and execution of seamanship and shiphandling to effectively control warship in typical operational conditions	<u>Activity# 1: Eight Seamanship Worksheets and associated lectures, discussion (15 hours classroom time)</u> Complete worksheets and discuss in class <u>Activity# 2:</u> Stability Lectures and Worksheet (7.5 Hours) <u>Activity #3:</u> Shiphandling Case Studies (4 hours) <u>Activity #4:</u> Seamanship Whiteboards -Scenario discussion (3 Hours) <u>Activity#5: NABS Practice:</u> (20 NABS Days) • Basic Alongside • On and Off Jetty Sets • Twisting on a Headrope • Tow Approaches • Replenishment • Anchor Work • Alongside Scenarios	d.
			To be developed
	Why	**Resources**	
	Seamanship is a key component of warship command	**For student:** Classroom and NABS as applicable **For instructor:** Assessment checklist Classroom and NABS as applicable	

CEC 002C – Shiphandling

Overview	Objective(s)	Assessment Details		Lessons Identified
Media: **Method**: Simulation **Time:** For Each Run: 1 Hour Min Preparation, 1 Hour Execution 2 NABS Bridge Days **Environment**: NABS		This EC consists of two separate NABS Scenarios. Each run will require the student to choose between two available options (berths) to bring their ship alongside in a foreign port. The Student will be evaluated in the criteria they use to make their decision, the planning of the alongside, the instruction to the team, and the execution. **<u>Before the assessment:</u>** The Student will be provided with a basic scenario and sufficient time to prepare **<u>During the assessment:</u>** The student will present his plan, identify key factors, risks and mitigations, and will then execute the plan in NABS **<u>Assessment feedback:</u>** The Assessing Officer will provide written feedback and assessment. The Overall assessment for the CEC will be a summative assessment of the two runs.		
	Impact and Limitations	**Resources**		**To be developed**
	Impact: Thie EC will form part of the Officers Evaluation and will be reflected in the course report. **Limitations / Remarks**:	**For student:** ECPINS	4 NABS Bridge Days 4 NABS Controller days	

EO 002.04 Oversee Bridge Resource Management

<table>
<tr><th>Overview</th><th>Objective(s)</th><th colspan="2">Activity Details</th><th>Lessons Identified</th></tr>
<tr><td rowspan="3">Strategy:
Combination of student and instructor led depending on the activity

Media:
Laptop
Projector

Method:
Lecture, practical application, discussion and syndicate work

Time:
Lecture/discussion – 2hrs
Syndicate work – 4 hrs
Practical application – 10 ORCA sea days

Environment:
Classroom
Sea Phase</td><td>- Understand the effect of human factors in accidents using the principles of BRM
- Describe best practices in decision making, closed loop communication, distraction control, Bridge ergonomics, shared mental model, situational awareness, risk management
- Understand the impact of fatigue and stress
- Understand and apply the error chain
Apply BRM in a practical environment</td><td colspan="2"><u>Activity #1 – Lecture</u>
Instructor led lecture and discussion on elements of BRM

<u>Activity #2 – Case Studies</u>
Students work in syndicates to analyse and present case studies by applying the principles of the error chain. Instructor will provide formative feedback and will facilitate discussion at the end of every presentation.

<u>Activity #3 – BRM Best Practices Development</u>
Concurrent with NABS runs for CEC 002A and CEC 002C

<u>Activity #4 – Practical Application</u>
BRM principles and best practices are applied continuously at sea during the ORCA sea phase.</td><td></td></tr>
<tr><td></td><td colspan="2">Why</td><td>To be developed</td></tr>
<tr><td></td><td colspan="2">Resources</td><td></td></tr>
<tr><td></td><td>Understanding human factors is critical to preventing accidents at sea</td><td>For student:
DWAN laptop with internet access</td><td>For instructor:
BRM lecture and case studies available</td><td></td></tr>
</table>

PC 002D – ORCA OIC Certification

Overview	Objective(s)	Assessment Details		Lessons Identified
Media: WECDIS Laptop Ship at sea **Method**: Practical **Time:** 10 Sea Days **Environment**: At Sea	To confirm the student's ability to competently operate as OIC of an ORCA Class PCTU	In accordance with the ORCA OIC Certification process, in accordance with the standard for ORCA OIC Certification. This PC focuses on basic platform knowledge and proficiency, with following caveat: The CDC Assessor may make this assessment as an ongoing effort of the overall two- week programme and need not conduct a series of dedicated check-rides as required by the PCTU QSP.		
	Impact and Limitations	**Resources**		**To be developed**
	Impact: **Limitations / Remarks**:	**For student:**	**Two ORCA PCTs and crew.** **- Two Assessor/Mentors who will act as OICs and should be current or recent warship COs.** **For instructor:** - Run sheet - Assessment Checklist	- Run sheet - Assessment Checklist

EO 003.01 Direct Tactical Level Mission Planning

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Combination of student and instructor led depending on the activity **Media**: Classroom with projector **Method**: **Time**: Lecture x 60min Practical exercise x 120min Guest Speaker x 120 min **Environment**: Classroom	1. Develop understanding of CO's role in tactical level planning and apply tactical planning processes to tactical problems; 2. Conduct Commanding Officer's personal estimate; and 3. Explain CO's role in course of action development.	<u>Activity#1: Lecture</u> Instructor led; highlights difference between tactical planning as a CO and tactical planning as an ORO. Introduces several different tactical planning processes; Students encouraged to come up with their own format. <u>Activity#2: Guest Lecturer</u> Current or serving CO engages in discussion about pre-deployment preparations <u>Activity#3: Planning Activity – Red Harbour</u> Students are broken down into syndicates and present plans for a tactical level exercise	
			To be developed
	Why	**Resources**	
	CO's role in tactical level planning is different from an OROs role	**For student:** DWAN laptop **For instructor:** a. Classroom b. projector	

EO 003.03 Explain the Conduct of Domestic and Expeditionary Operations

Overview	Objective(s)	Activity Details	Lessons Identified
Strategy: Combination of student and instructor led depending on the activity **Media**: Worksheet discussion x 60min Guest Lecturer x 120min **Method**: **Time:**	1. Explain federal maritime jurisdictions in Canada: a. Public Safety and maritime emergency response plan; b. RCMP; c. Transport Canada; d. Department of Fisheries and Oceans; e. Canadian Border Service Agency; and f. Canadian Coast Guard. 2. Explain the conduct of missions in support of Other Government Departments. 3. Explain CONPLANS	<u>Activity #1 – Worksheet</u> worksheet to be completed as homework and then discussed in class <u>Activity #2 – Guest Lecturer</u> Usually a senior member of the MARPAC N3 staff to discuss the command and control relationships in both expeditionary and domestic operations	
Environment: Classroom			**To be developed** Worksheet

Why	Resources	
	For student: Access to reference library DWAN laptop	**For instructor:**

EC 003.01 – Maritime Law

Overview	Objective(s)	Assessment Detail		Lessons Identified
Media: Online **Method**: TEAMS Worksheet and subsequent discussion **Time**: 4 hours classroom time **Environment**: Online / Classroom	To confirm the Students command level Knowledge of relevant Maritime Law	This EC is conducted as an open book directed publication search, asking questions of the student, some of which have clear answers, others which are somewhat subject to interpretation. The Student is required to find or state their answer and to provide the specific reference which supports their finding. A Legal Officer is then invited to partake in the post worksheet discussion of the questions raised.		
	Why	**Resources**		**To be developed**
	- Understanding Maritime Law is a core aspect of sea command.	**For student:** Access to reference library in particular: • UNCLOS • San Remo Manual DWAN laptop	**Command Experienced Instructor** **Legal Officer**	

EC 003.02 – Use of Force

Overview	Objective(s)	Assessment Detail		Lessons Identified
Media: Online **Method**: TEAMS Worksheet and subsequent discussion **Time**: 4 hours classroom time **Environment**: Online / Classroom	To confirm the Students command level Knowledge of subjects pertaining to the use of force in Naval Operations	This EC is conducted as an open book directed publication search, asking questions of the student, some of which have clear answers, others which are somewhat subject to interpretation. The Student is required to find or state their answer and to provide the specific reference which supports their finding. Topics to be Included are: The Relationship between International Humans Rights Law and the Law of Armed Conflict Standard of Care for Accountability in Use of Force Self Defence, Individual and National, Limiting, Others that may be defended without ROE etc., Defence of Property Determination of Hostile Intent Application of Warnings in the Marine Environment Canadian Force Participation in Domestic Operations and Support to Law Enforcement A Legal Officer is then invited to partake in the post worksheet discussion of the questions raised.		
	Why	**Resources**		**To be developed**
	The Commanding Officer must be an expert in the interpretation of the CF Use of Force Manual and its application to use of force in the Marine Environment	**For student:** e. Access to reference library in particular the CF Use of Force Manual f. DWAN laptop	**Command Experienced Instructor** **Legal Officer**	

Part 3 - Assessment Checklists

CDC Shiphandling Enabling Check Assessment
EC 203.01

Candidate:	**Assessor:**	**Date:**
		EC Run # /2

Objective		**Dev/Sat/Strong / N/A**	**Comments**
Choice of the Alongside Berth	Considers relevant factors (wind/ tide / available room/exit)		
	Appreciates Risk and mitigations options		
Discussion of the Plan. Correctly understands or instructs:	Environment		
	Execution		
	Risks & Mitigations		
	CDP/ PNR (Only if and as appropriate)		
	Line Handling		
	Direction to Team		
Execution of the plan. Correctly applies:		**Dev/Sat/Strong / N/A**	
	Speed Regime		
	Assesses and reacts to set		
	Line of Approach		
	Rudder		
	Engines		
	Appropriate Use of ECPINS, Eye, Pelorus		
	Direction to Team		
Overall Assessment & General Comments **Developing – Did not meet standard** **Satisfactory – Meets standard** **Strong - Exceeds standard**			

Assessor Signature: _____

ORCA Sea Phase PC 001/PC 002D

Name:	Mentor:	Date:
Summary Of Activities:		

Performance Check	Assessment Criteria
PC001 Command Decision Making ☒ = SAT ☐ = UNSAT	☐ Wrote coherent, well-considered night orders ☐ Proactively identified when to be on the Bridge ☐ Responded appropriately to calls from the OOW ☐ Planned when and where to conduct training activities given risk factors. ☐ Managed consort deliberately ☐ Made Routing and programmatic decisions based on unforeseen challenges ☐ Considered whether to continue with a seamanship event given escalating risk factors ☐ Responded to engineering casualties and difficulties; ☐ Made decisions regarding crew members; ☐ Responded to changing weather; ☐ Responded proactively to SAR notifications (exercise or real-time) ☐ Maintained situational awareness **Leadership** ☐ Clearly communicated intent to the team ☐ Motivation and energy level – proactive/reactive? ☐ Able to maintain composure when under pressure
Comments	Comments to be provided by assessor on strengths/weaknesses in any area. Attach supplemental sheets should you have additional comments to provide. [illegible]
PC002D OIC Assessment ☒ = SAT ☐ = UNSAT	☐ Shiphandling (departures and alongsides) ☐ Pilotage, anchoring and entering/leaving harbour ☐ Damage Control exercise ☐ Electrical blackout or Steering Gear breakdown ☐ Log strike ☐ Man overboard ☐ Engineering difficulty/emergency ☐ Response to loss of GPS, or other navigation fault.

	Comments to be provided by assessor on strengths/weaknesses in any area. Attach supplemental sheets should you have additional comments to provide.
Comments:	

Explanation of Assessment Criteria/Rubric

IAW the CDC QSP version xxx, updated fall 2020; two (2) PCs are being assessed during the ORCA Sea phase:

PC 001 – is a subjective assessment of the officer's competence in exercising decision making, risk appreciation and management commensurate with the commanding of an ORCA class vessel in typical multi-ship day/night operations. The Student will spend up to 24 hours as the Captain and will be assessed on the quality of their decision making and risk management as it pertains to the following common activities. It should be noted that students will be evaluated on their decisions, risk assessment and leadership throughout their time at sea, not just when acting as OIC. When considering the student's appreciation of risk it is important to remember that the mission is to maximize training opportunities in the time allotted. Students should be looking for ways to achieve the mission and should only cancel serials or activities in the unlikely scenario that a creative solution cannot be found.

PC 002D – is the ORCA OIC at sea assessment checklist. Every student must complete the items below to the satisfaction of the Assessor. Re-attempts are permitted. The list provided on the assessment table is taken from the ORCA OIC assessment checklist which is available on the PCTU website.

Element Under Assessment		Criteria for SAT to be awarded
Planning and Execution Criteria	**Night Orders**	- Provided in a timely fashion and discussed with the OOW - Risks identified and mitigations put in place (ie. CO to be called) - Orders as written will result in aim being achieved **Note**: OOW/XO should not be delegated any of OIC's authority
	On the Bridge	- Maintain Situational awareness - Identify risk (s) (traffic, weather, engineering issues) - Support the OOW appropriately when/if overwhelmed
	Responded appropriately to calls from OOW	- Identify risk based on description of the situation (or lack thereof) - Able to rouse and be alert during night hours
	Planned where/when to conduct training activities	- Applied critical thinking to serial planning in relation to busy traffic areas - Adapted serial to account for unexpected traffic/weather concerns or other challenges - Developed alternative solutions when/where required, so as to preserve training serials
	Managed Consort Deliberately	- Give clear instructions to the OOW in the management of consort - Get ahead and prepare the majority of signals to be executed by leveraging the delayed executive method - Incorporate the consort in anti-collision manoeuvring decisions
	Routing and Programmatic Changes made based on Challenges presented	- Proactive vs reactive - Solutions sought so that mission could still be achieved - Risk considered when routing choices made
	Considered whether to continue with seamanship events	- Considered crew fatigue, weather and engineering/technical challenges
	Responded to engineering challenges	- Applied MBS - Considered risk, reward and mission when making decisions - Looked for creative solutions (hull swap etc) -

	Responded to changing weather	- Maintained awareness of current weather and forecasted conditions - Was aware of restrictions imposed upon ORCA class - Confirmed equipment accuracy by employing transits
	Responded proactively to SAR	- Determines proximity to stricken vessel and responds if appropriate - Contact JRCC - Oversee tactical planning process
Leadership	**Clearly communicated intent**	- Clearly communicate during emergencies - Establish priorities
	Motivation	- Positive energy - Looks for opportunities - Proactive vs reactive
	Ability to maintain composure under pressure	- Speaks respectfully to others when things aren't going well - Gets along with team mates - Does not blame others

Instructions:

1. Place a check mark next to those activities which the student has completed to satisfactorily.
2. If the student did not met the requirement please comment on why they were not able to do so.
3. If a student is particularly strong in an area, please comment on how this was demonstrated.
4. A min of one day prior to the end of the sea phase, provide a short narrative paragraph to CDC staff, indicating your final assessment.
5. If necessary, please keep divisional style notes on students on a daily basis to justify assessments.

CEC 002.01 Collision Avoidance

Candidate:	Assessor:	Date:
Scenario:	CEC Serial # /4	

Assessment Criteria	Comments				
Uses Available time to appreciate a developing collision situation – "Took the watch"	Time Arrived on Bridge		Nav Lights? Yes ☐ No ☐ N/A ☐	AIS? Yes ☐ No ☐ N/A ☐	
	Begins to Implement a plan		Sound Sigs? Yes ☐ No ☐ N/A ☐	Radar Ranges OK? Yes ☐ No ☐ N/A ☐	
	Time spent assessing:		Eng Config? Yes ☐ No ☐ N/A ☐		
Develops a suitable plan					
Executes the plan Effectively					
Uses Correct Tactical Manoeuvering Procedures					
Manages Bridge Resources Effectively					
Overall Assessment & General Comments	1. Developing – Needs more practice 2. Satisfactory				
Assessor Signature					

EC 002.01 Navigation

Candidate:	Assessor:	Date:
Scenario:		

Assessment Criteria	Comments

Bridge Set-Up
• SDM
  ○ Feature set - Coastal pilotage (day)
  ○ Correct chart type
  ○ Correct position source selected
  ○ Safety depth
  ○ Radar overlay displayed and accurate
  ○ Anti-grounding cone displayed (optional)
• Radar
  ○ Relative motion vectors, true trails
  ○ Appropriate range scale
  ○ IR on
  ○ Contacts selected
• AIS checked and updated
  ○ Ownship data correct
• Echo sounder set-up
Equipment errors mitigated	

Basic Planning • Track selection • Track scan conducted ○ Using XTC or 1:1 visual scan • Speed plan • High risk areas identified and mitigated Risk Management • Traffic • environmental effects • navigation hazards, equipment degradation prioritization	

Execution
• Use of transits or visual bearings to determine ship's position/safety
• Participated in VTMS
• Radar overlay checked
• Responds appropriately to equipment faults
  o Identifies faults
  o Mitigates with back up systems (visual fixing, PI/SI, O/C bearings etc
• Conning
• Prioritization
• Application of COLREGs
  o Contact avoidance
  o Lights and sound signals
Situational Awareness	

Overall Assessment & General Comments	1. Developing – Needs more practice 2. Satisfactory
Assessor Signature	

CEC 002C Ship Handling

Candidate:	Assessor:	Date:
		EC Run # /2

Objective		Dev/Sat/Strong / N/A	Comments
Choice of the Alongside Berth	Considers relevant factors (wind/ tide / available room/ exit)		
	Appreciates Risk and mitigations options		
Discussion of the Plan. Correctly understands or instructs:	Environment		
	Execution		
	Risks & Mitigations		
	CDP/ PNR (Only if and as appropriate)		
	Line Handling		
	Direction to Team		
Execution of the plan. Correctly applies:	Speed Regime		
	Assesses and reacts to set		
	Line of Approach		
	Rudder		
	Engines		
	Appropriate Use of ECPINS, Eye, Pelorus		
	Direction to Team		

Overall Assessment & General Comments 1. Developing – Needs more practice 2. Satisfactory 3. Strong	
Assessor Signature	